ETHAN OLVERA

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Portfolio:https://sites.google.com/view/ejrolvera/home | GitHub:github.com/TheRealET06

EDUCATION

University of Illinois Urbana Champaign — Urbana Champaign, IL | **GPA : 4.0**
(Transfer from Northern Illinois University)
Bachelor of Science – Computer Science | Expected Graduation: Spring 2028

TECHNICAL SKILLS

Languages: C++, Python, Go, Java, JavaScript, C#, Assembly, HTML, CSS, PHP, Bash,JCL
Technologies: Kubernetes, Docker, Unreal Engine, OpenXR, OpenCV, TensorFlow, Git, Mainframe, IBM System/370
Tools: Optuna, Prometheus, wrk2, LIV.tv, Weight & Biases, DaVinci Resolve
Creative / Media: Video Editing, Digital Content Production, Social Media Management

TECHNICAL PROJECTS

Google Halloween 2016 Doodle – Computer Vision / Machine Learning

- Developed a **computer vision and machine learning system** to recreate the Google Halloween 2016 Doodle game.
- Built a custom **TensorFlow model** to interpret hand-drawn symbols in the air and map them to in-game inputs.
- Used **OpenCV and Pygame** to capture, process, and display players' air-drawn inputs in real time.
- Integrated the machine learning model with the **real-time computer vision pipeline** to translate player movements into interactive game controls.

EXPERIENCE

Undergraduate Researcher – NSF REU, Cosimo Project
Illinois Institute of Technology | Chicago, IL | May 2026 – August 2026

- Developed **Cosimo**, a co-simulation framework for optimizing Kubernetes container orchestration using **Go and the Kubernetes API**.
- Implemented a closed-loop simulation workflow that **monitored Kubernetes cluster state, adjusted workloads, and evaluated system performance** under varying load conditions.
- Deployed and benchmarked the **DeathStarBench Social Network** microservice application on Kubernetes to evaluate scalability and resource management.
- Developed workload experiments using **wrk2** and analyzed throughput, latency, CPU utilization, and replica scaling behavior under increasing request rates.
- Applied **Optuna hyperparameter optimization** to identify Kubernetes autoscaling configurations that maximized sustainable application throughput.
- Integrated **Prometheus metrics and custom synthetic-load metrics** to provide performance data for Kubernetes autoscaling and co-simulation experiments

Undergraduate Researcher – Virtual Production / XR
Northern Illinois University Office of Innovation | DeKalb, IL | September 2025 – May 2026

- Led development of an **XR virtual production studio**, integrating Unreal Engine, OpenXR, LIV.tv, VR hardware, and real-time rendering pipelines.
- **Earned 2nd Place at the 2026 NIU CURE Research Conference** in the Computer Science, Physical Science & Mathematics category.
- Engineered **real-time mixed-reality capture workflows** using LIV.tv, motion tracking, and VR hardware for immersive research applications.
- Developed and tested **Unreal Engine XR prototypes**, evaluating performance, usability, and interaction workflows.
- Collaborated with researchers across disciplines to design and implement **interactive XR applications and virtual production pipelines.**

Student Program Development Lead
Northern Illinois University Office of Innovation | DeKalb, IL | September 2025 – December 2025

- Led the creation and development of NIU's **Computer Science Hackathon Team**, establishing a collaborative and entrepreneurial team environment.
- Led teams in **national and international hackathons**, developing innovative solutions under tight deadlines and competitive constraints.
- Applied **analytical problem-solving and rapid decision-making** to design, prototype, and refine hackathon solutions.
- Mentored students in **collaborative software development, experimentation, and solution design**, strengthening technical and problem-solving skills.

System & Web Administrator (Volunteer)
American Legion Illinois Premier Boys State | Charleston, IL | June 2025 – Present

- Oversaw daily operations and troubleshooting of **email systems and websites**, maintaining reliable digital services.
- Diagnosed and resolved **technical issues affecting system performance and reliability.**
- Collaborated with media and operations teams to deploy **digital platforms and communication tools** for program participants.